AirTransformer Resilience

Climate resilient cooling for urban critical facilities

Savery Tech · France · Sole beneficiary

HORIZON-EIC-2026-ACCELERATOR-02-CHALLENGES-05

Deep Tech for Climate Adaptation · Blended finance · 24-month grant action

Grant request €2,485,583.33 · EIC investment request €2,000,000 · Total EIC request €4,485,583.33

Eligible action cost €3,550,833.33. Non-grant eligible share €1,065,250. The €2m investment request forms part of the company's €8m Seed target, alongside €6m sought from other sources.

Executive summary

AirTransformer provides cooling for facilities that must keep operating through urban heatwaves, electricity constraints and water stress. Savery Tech already has a complete TRL6 machine combining sorption air treatment, thermal management and indirect evaporative cooling. Its CAU-series material route uses low-temperature regeneration heat, with a product qualification envelope of 50–70°C heat supply, 16–20°C delivered air and a humidity ratio of 9–11 g/kg dry air under the applicable rating conditions. The productisation targets are 60–70% lower electricity demand and 40–60% lower external make-up water at equivalent cooling service; installation economics are screened against a 24–36-month payback target.

Savery owns its 2019 system patent family and controls the product architecture and engineering know-how. Its scientific board includes Norbert Stock and Aline Auroux. An awarded CETP grant supports the ongoing 2026–2029 project towards TRL7, with HPE providing the site. The EIC action builds on this basis to qualify repeatable manufacture, reliability, compliance and deployment at TRL8. The 24-month programme requests €2,485,583.33 grant against €3,550,833.33 eligible costs, plus €2m EIC investment within an €8m Seed round. The investment supports co-financing and commercial deployment, while the grant buys the distinct product evidence required for a repeatable European cooling business.

1 Technology

1.1 Novelty and breakthrough nature

A working cooling product with a defined engineering advance

AirTransformer is a modular cooling and air-treatment product for facilities with a continuous heat load and access to suitable low-grade heat. Its first commercial focus is data-centre infrastructure, where high ambient temperatures can coincide with electricity-network constraints and restrictions on cooling-water supply. The product combines moisture removal by metal-organic framework sorbents with thermal management and indirect evaporative sensible cooling. It supplies conditioned air at the agreed delivery interface while the facility retains its existing cooling installation for redundancy and duties outside the module's operating envelope.

Savery has already integrated the core functions in a complete TRL6 machine. The present innovation effort concerns making this product dependable, manufacturable and economical across qualified installations. It addresses the engineering relationship between the CAU-series sorbent, its shaped and coated heat exchanger, the cyclic thermal process and the customer's required service. A material with attractive equilibrium uptake is useful only if the manufactured contactor can capture and release moisture quickly enough, with acceptable pressure loss, heat transfer and ageing. Similarly, an efficient contactor does not establish an efficient installation if regeneration, pumping or heat rejection consumes the benefit.

The breakthrough proposition is to deliver the required air service using a thermal resource that many facilities already reject, while reducing demand for electricity and external make-up water. Its defensible value lies in the complete installation: air-side performance, utility interfaces, repeatable production, maintenance and acceptance evidence. The action therefore advances an existing integrated machine through product engineering and qualification rather than beginning with an open search for a new sorbent chemistry. [T01, T02]

How the system delivers cooling

The process-air path first passes through the sorption section. The CAU-series material captures water vapour and the thermal circuit removes the heat released by adsorption. Drying lowers the air's wet-bulb temperature and creates a more favourable driving condition for the downstream indirect evaporative section. In that section, water evaporates into a separate working-air stream and removes sensible heat across the exchanger. The cooled product-air stream remains separate from the humid exhaust, so evaporated water is not added to the delivered air.

The contactors alternate between treatment and regeneration. Low-grade heat restores the sorbent's working capacity, while switching and heat recovery coordinate the availability of the next contactor. Regeneration air, product air and working air have separate interfaces. The thermal design must accommodate both adsorption heat and the regeneration requirement; rejected heat must reach a compatible sink. Fans, pumps, actuators, controls and water treatment all contribute to the installation's electrical demand. These loads are included when comparing alternatives, together with any residual duty supplied by the retained plant.

The product's water benefit starts with reduced external make-up for the same delivered cooling. Atmospheric moisture that is recovered as liquid can contribute to that balance when the regeneration and condensation conditions allow it. The water ledger distinguishes capture, condensation, treatment losses, storage changes, discharge, internal consumption and net usable output. Recirculating an existing water inventory is not counted repeatedly as new production. Recovered liquid is qualified for its actual use, with isolated non-aerosol equipment cleaning as a possible initial application. The action makes no fixed daily water-output promise independent of humidity, heat-source availability and heat-sink conditions.

This arrangement gives the facility an additional operating choice. During heatwaves, the operator can allocate a defined air-treatment duty to AirTransformer while preserving incumbent cooling for peak or exceptional conditions. When available heat, humidity or sink temperature moves outside the qualified domain, the controller changes mode or bypasses the module. Cooling continuity therefore depends on a specified allocation and fallback arrangement, supported by a measurable service contract and commissioning test.

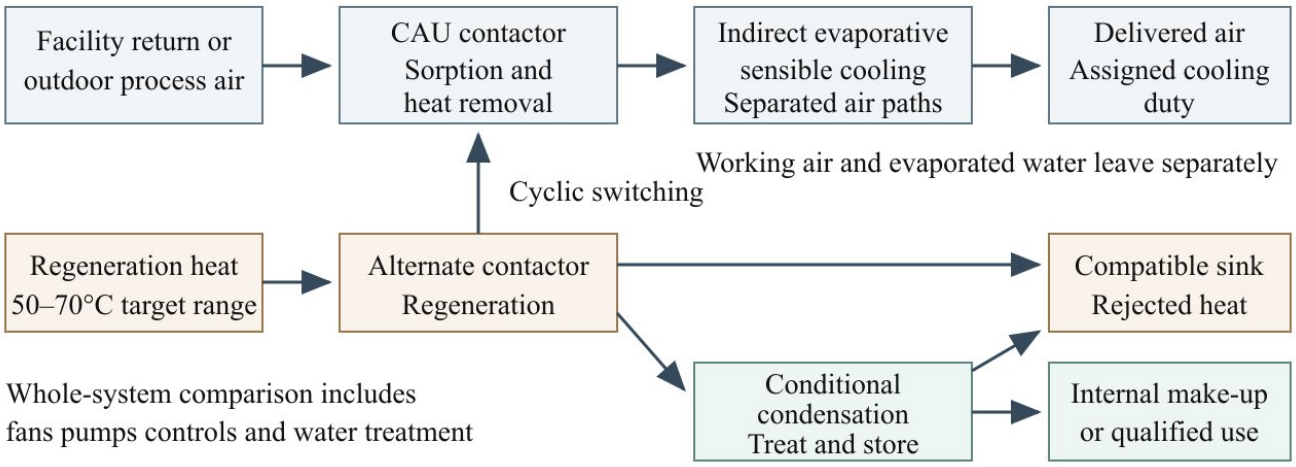


Figure 1. Product service paths and site interfaces. Separate air streams preserve the delivered-air boundary. Regeneration heat and rejected heat have distinct interfaces. Liquid recovery depends on compatible condensation conditions and use qualification.

CAU material processing and contactor manufacture

The current engineering material route uses CAU-series variants. Qualification will establish a controlled synthesis and conditioning window, preserve the adsorption response after shaping, and transfer the process to repeatable contactor manufacture. The relevant outputs are the material and process specification, an accepted batch, a reproducible coating or shaped structure and a tested replacement procedure. Broad chemical screening is not the starting point of the action.

Processing determines how much of the material's capacity remains accessible in the machine. Binder content, coating thickness, porosity, adhesion and thermal contact influence moisture-transfer rate, pressure loss and cycle duration. The process-development work will therefore measure performance after shaping and assembly, rather than release powder solely on an equilibrium isotherm. Incoming-material identity, preparation conditions, activation, substrate preparation and final conditioning will be traceable. Manufacturing acceptance will connect these records to adsorption response, airflow resistance, coating integrity and thermal performance.

The contactor also has to survive its intended operating environment. Condensation, particulate exposure, contamination and cleaning can change the surface and transfer behaviour. Qualification specimens and assembled units will undergo representative exposure and recovery sequences, followed by the same acceptance measurements used at release. Where performance changes materially, the response will be a revised operating or maintenance limit, process adjustment or replacement criterion. This creates a product quality system that can distinguish a manufacturing deviation from normal ageing and can support service decisions in the field.

Savery will use existing facilities where they meet the task. Additional equipment, specialist testing and external processing will be selected by their contribution to process control, product evidence and delivery capacity. The industrialisation output is a transferable manufacturing package comprising drawings, bills of material, tolerances, process instructions, inspection points, test methods and release criteria. Critical process knowledge remains under Savery's control, while qualified suppliers receive the information and permissions necessary for their assigned work.

Finite-time design and measured transport

Savery's design capability connects sorption equilibrium to the time available in a working machine. The useful quantity is the moisture exchanged during a complete repeatable cycle, at the actual source and sink conditions, together with the cooling delivered during that time. A long adsorption tail may add little service while delaying regeneration; an early switch can increase cycle frequency but leave capacity unused. Heat recovery also trades reduced heating demand against the time needed to prepare the next contactor.

Published work co-authored by Frédéric S. Cui supports this engineering approach. Henninger et al. used large-temperature-jump experiments on CAU-10-H and aluminium fumarate to relate progress through adsorption to characteristic times and mass-, area- and volume-specific mean powers. The study shows why ranking materials by an isolated time constant can misrepresent useful power. Savery applies this understanding to shaped contactors, thermal interfaces and complete operating cycles. [T03]

Measured temperature and vapour-pressure responses constrain a dynamic model of loading, heat transfer and flow resistance. Parameter identification is directed at engineering decisions: selecting coating and contact conditions, choosing switching times, and recognising when an aged or contaminated contactor needs a different operating mode. The model retains its tested range and measurement uncertainty. A fitted parameter is useful only when the available measurements distinguish it sufficiently to inform those decisions.

Thermodynamic length provides a local description of dissipation where the measured system supports a linear-response approximation. It helps identify paths that avoid unnecessary gradients for a specified transfer and cycle duration. Larger excursions, hysteresis and switching retain their calibrated nonlinear description. The action applies these methods to product optimisation; it does not depend on creating a universal thermodynamic theory. [T04]

Performance comparisons close the complete-cycle balances and retain comparable initial and final sorbent loading and thermal inventories. This prevents a favourable fragment of operation from shifting regeneration or restoration beyond the evaluation period. Delivered cooling, electrical demand, regeneration heat and external water remain separate measures. The resulting design decisions can then be tested directly on the manufactured equipment.

Control that preserves service and safe fallback

The existing machine provides the baseline for cyclic operation and thermal management. The product upgrade will develop deterministic supervisory control and model-predictive scheduling, with a bounded reinforcement-learning layer evaluated for setpoints and cycle strategies where it adds useful adaptation. The learning task concerns decisions whose consequences extend across cycles, such as balancing additional adsorption against subsequent regeneration time. Physical limits, safety interlocks, abnormal-condition detection and fallback remain outside the learned policy. Rules and MPC provide qualified alternatives under uncertain state estimates, sensor faults or conditions outside the training domain. The verification plan uses reserved operating histories, equal sensor and actuator access, deliberate disturbances and recorded fallback behaviour. Human operators retain mode selection, alarms, override and safe shutdown. These upgraded predictive and learned functions are development tasks, distinct from the current TRL6 machine's already implemented controls.

A quantified product specification and transparent comparison

The qualification specification restores the concrete service proposition. It defines a 50–70°C regeneration-heat range and delivered-air targets of 16–20°C with 9–11 g/kg dry-air humidity ratio at applicable rating conditions. Humidity ratio is a water-content measure, not a dew-point temperature. Source temperature, flow, inlet humidity, load and sink condition accompany each rating point. The whole-machine result is calculated at net product-air delivery after working-air diversion and includes the associated auxiliaries.

The electricity and water targets are stated at equivalent useful cooling, with a technically capable reference installation. They will be assessed across the operating envelope rather than inferred from a single favourable component test.

Measure	Product or commercial objective	Basis for qualification
Regeneration heat	50–70°C within the selected operating domain	Measure source, core and return temperatures, flow, duration and available heat; distinguish source supply from material temperature
Delivered air	16–20°C and 9–11 g/kg dry air at qualified rating points	Record net airflow, temperature and humidity ratio together with inlet and sink conditions
Cooling electricity	60–70% reduction as a productisation target	Compare total module and affected plant electricity at the same cooling service, including heat-rejection and treatment auxiliaries
External make-up water	40–60% reduction as a productisation target	Compare incoming external water at equal cooling, after inventory changes, blowdown, treatment and internal reuse
Customer payback	24–36 months for qualifying installations	Use installed cost, operating hours, electricity and heat prices, external water and incremental maintenance
Hardware selling-price planning basis	€1,100/kW pilot-series hardware selling-price assumption, progressing towards €900/kW with repeat production	Keep hardware, installation, service costs and cooling-power definitions distinct; validate the scale assumptions through supplier and manufacturing evidence

The electricity target is consistent with an illustrative equal-service calculation. For a 1 MW cooling duty, a compressor reference at electrical EER 3 draws about 333 kW. A complete module at electrical EER 8–10 draws 125–100 kW, implying a 62.5–70% reduction before any additional installation duty. Qualification will test the complete installed boundary, while reporting the regeneration heat separately. This calculation explains the target; it is not presented as an already measured field reduction.

Likewise, a 40–60% water-reduction target means using 0.6–0.4 m³ of external make-up for a service that requires 1 m³ in the reference system. It does not mean that the module produces 0.4–0.6 m³ of recovered water. The baseline must include the actual cooling configuration, local climate and water-management practice. The commercial analysis uses these measured or modelled service quantities and site tariffs, with lower utilisation and heat availability tested explicitly. The action will replace planning values with accepted product and site results as qualification progresses.

Competitive advance and timing

Capable alternatives include efficient compressor systems, indirect evaporative equipment, desiccant treatment and heat-recovery systems. Their performance depends on climate, load, water availability and installation constraints. AirTransformer's intended advantage is the combination of low-temperature thermal input, controlled air-side cooling and lower external water demand in a retrofit arrangement that preserves the facility's existing plant.

Low-temperature heat alone is not a unique market distinction. AirJoule's disclosed Prime and Core product families use sorbent cycles, and its Core DH announcement identifies regeneration at 60–70°C. AirJoule also combines water and cooling activities with BitSink. Savery will therefore compare delivered function, air and water interfaces, the complete electrical and thermal boundary, footprint, installation, maintenance and cost. Water-production data from another product cannot be converted into a cooling-energy comparison without a common service definition. [T05]

The timing is driven by the conjunction of an existing TRL6 machine and an adoption problem that needs product evidence. Operators can assess a retrofit against a real cooling duty; manufacturers can qualify drawings and process windows; independent laboratories can test a released configuration. The EIC action connects these steps so that the commercial decision rests on repeatable equipment and accepted service, rather than a further laboratory demonstration of the underlying sorption principle.

1.2 Technology readiness level

Current complete machine at TRL6

As of November 2026, Savery has a complete TRL6 AirTransformer machine. Its existing technical basis combines staged MOF-coated heat-and-mass exchange, adsorption/regeneration switching, water-side heat management, indirect evaporative sensible cooling, control functions and an integrated measurement approach. This is the starting product for the EIC action. The current material engineering route is CAU-series variants. [T01, T02]

The development record includes integrated laboratory operation and testing of air treatment under controlled thermal and humidity conditions. The engineering evidence has progressed from material response and exchanger behaviour to combined air, heat and moisture balances. The platform evidence map dated 5 August 2026 records the integrated prototype and relevant-condition engineering basis. Subsequent current company confirmation establishes the completed-machine TRL6 status used in this application. The EIC programme will qualify the released product configuration, preserve a traceable baseline, and extend its evidence to repeat manufacture and operational acceptance. [T01, T06]

Current maturity is kept separate from the proposed upgrades. New coating windows, supplier transfer, reliability qualification, predictive control and the optional learning layer require their own verification. Additional liquid recovery and water treatment will be qualified for the actual use. These development tasks build on the complete machine; their future status does not turn established sorption, switching or thermal-management functions back into early-stage research.

Product element	Existing basis	EIC productisation increment	Acceptance evidence
Complete air-treatment machine	Integrated TRL6 configuration with sorption, thermal management, cyclic switching and indirect sensible cooling	Freeze the product design and operating modes, define service interfaces and qualify the installed boundary	Released drawings and control baseline; repeatable performance map and customer-facing operating envelope
CAU material and contactor	Current CAU-series route and coated-exchanger development	Process windows, after-shaping performance, supplier transfer and batch consistency	Traceable lots; incoming and final acceptance results; deviation and replacement rules
Cyclic control	Existing machine control and switching functions	Deterministic supervision, predictive scheduling and bounded learning evaluation	Fault injection, disturbance recovery, reserved-condition tests and safe fallback records
Reliability and service	Existing prototype and component experience	Ageing, contamination, cleaning, maintainability and repeated operation	Life-test results with inspection intervals, failure analysis and service procedures
Compliance and product release	Existing engineering baseline and applicable-method assessment	Independent product testing, technical file and installation acceptance	Test reports for the released configuration, conformity evidence and documented acceptance
Deployment	Awarded CETP programme with HPE site towards TRL7	Distinct manufacturing and product qualification, site engineering and deployment replication towards TRL8	Incremental operational acceptance and a deployable manufacturing, installation and service package

CETP and HPE provide an established validation pathway

Savery has received a CETP grant and the 2026–2029 project is underway, targeting TRL7 with a site provided by HPE. This is an existing collaboration and validation foundation. It provides access to relevant integration knowledge and a route to operation under data-centre conditions. HPE's role in that programme is distinct from a new purchase order or an additional EIC site commitment. [T01]

The programmes have complementary results. CETP addresses the technical integration and operational validation pathway, including materials and kinetics, heat and mass transfer, cycle maps, digital modelling and control, and HPE integration. The EIC action concentrates on the work that makes a reproducible commercial product: design for manufacture and assembly, controlled processes and batches, durability and maintainability, applicable compliance, independent product tests, acceptance and deployment replication.

The EIC schedule runs from April 2027 to March 2029. It can use CETP outputs that are available at the relevant design or test gate, but it does not rely on a late-2029 result to complete in March 2029. Design and operating baseline work begins from the current TRL6 machine. Product qualification can use an independent or equivalent relevant-condition test route when a shared site window or later CETP activity is unavailable. Any result adopted from CETP carries its configuration, operating conditions and limits into the EIC evidence dossier.

Financial and technical responsibility follow the same boundary. The same employee time, equipment use or experiment is charged once. Shared background data and earlier results can be reused, while the EIC cost plan covers its separately defined productisation tasks. Site interfaces and measurement arrangements can be coordinated without treating the CETP grant as co-financing for the same EIC cost. The project manager maintains the activity and cost allocation, and technical change control records when a shared result affects the released product.

From TRL6 to a qualified and reproducible product

The first gate establishes the current machine's product baseline: configuration, intended use, service boundary, operating domain and the measurement and acceptance plan. Existing records will be organised by configuration so that a later process or controller change can be tested against the correct baseline. The design review sets the remaining material, mechanical, controls and supplier tasks and gives each a responsible engineer and budgeted resource.

The manufacturing gate qualifies the chosen CAU processing and contactor route, assembly sequence and inspection plan. It tests independent preparations and batches against product-relevant acceptance criteria. This gate also establishes which measurements can be performed economically on every unit and which require periodic or sampled verification. Manufacturing release requires reproducibility across accepted variation, not just the best result from one specimen.

The reliability and conformity gate tests the released system under representative thermal, humidity, load and contamination changes. Long-run behaviour, switching transients, abnormal conditions and maintainability are assessed with the complete service boundary. A component change that affects performance, safety or water use triggers the relevant retest. Product claims will identify the configuration, conditions and uncertainty of the result rather than extend a material or component result to every installation.

Final product acceptance brings together the qualified hardware, control and manufacturing versions with the installation and service package. The TRL8 objective is a complete system qualified for its intended use, with production and operational evidence sufficient to support repeat deployment. Customer acceptance will concern useful cooling continuity, agreed air conditions, utility use, alarms and fallback, maintenance access and the responsibilities needed for safe operation. The incumbent system remains available throughout commissioning and relevant trials.

Testing will use ASHRAE 143-2024 for the applicable steady-state indirect evaporative measurements, supplemented by a protocol for the complete cyclic machine. The metrology plan will assess relevant air-side procedures from EN 13141-7, recognising its residential ventilation-unit scope and documenting any adaptations. Calibration will use laboratories with the relevant ISO/IEC 17025 competence and measurement traceability. Method use and traceable calibration will be distinguished from certification of the entire machine. The conformity work will determine the applicable product legislation and tests for the actual installation and electrical equipment; the technical file, risk assessment and declaration follow the released design. [T13]

Climate resilience is an explicit acceptance question. The test matrix will combine hot conditions with constrained electrical and water resources and will record the service that the product retains under those constraints. It also includes low heat availability, unfavourable humidity and warm sinks. Where the product cannot supply the full assigned duty, its permitted operating modes and the retained-plant response form part of the installation design. This makes the resilience claim measurable and gives operators a practical basis for using the technology during the conditions that matter most.

1.3 Intellectual property strategy

Savery's system rights and product know-how

Savery owns the system patent originating from the 2019 filing. The identifiable family includes European application EP19306516.6, published as EP3825627A1, and PCT/EP2020/082838, published as WO2021099546A1. These publications concern cascaded adsorption/desorption cooling and heating and provide the foundation for Savery's system-level protection. The current product uses the CAU-series route, with Savery controlling the architecture and engineering integration. [T01, T07]

Protection covers different kinds of assets. Patentable system and contactor developments are considered for filing before disclosure. Manufacturing recipes, coating and conditioning windows, tolerances, calibration methods and process-control knowledge are managed as confidential know-how. Software, models, firmware, test data and service tools are controlled through copyright, access rights, version management and agreements that preserve Savery's ability to manufacture, maintain and improve the product.

The 29 January 2025 Plasseraud analysis examined the patentability of a two-MOF heat-and-mass exchanger and related material use. It identified a route to system and application claims while requiring clearer evidence of the arrangement and technical benefits. Savery's subsequent product work will connect claim drafting to documented engineering effects and the released embodiment. A patentability analysis informs protection strategy; the commercial rights assessment additionally considers third-party claims. [T08]

The current action will create its own improvements to material processing, shaped contactors, control and manufacturing transfer. Employee, contractor and collaboration arrangements will identify background assets, ownership of results, permitted use, confidentiality and the rights needed for supply and service. External manufacturers will receive controlled process and quality instructions under appropriate use rights. Savery will retain the information and permissions required to qualify another supplier or maintain deployed equipment.

Product and territory specific third-party assessment

The freedom-to-operate review follows the actual product, from synthesis and shaping to assembled heat exchangers, thermal circuits, software and installation. Relevant countries are identified by activity: European manufacture and first deployments, followed by the countries selected for sales or contract production. A publication or a competitor's product name is a screening lead; the decision concerns applicable claim language, legal status and the proposed act in a particular country.

The existing professional work provides a focused starting point. The HMX analysis separated material claims, substrate and adsorption-cooling claims, indirect evaporative geometries and liquid-desiccant combinations. Product engineering will preserve that distinction. Changing to CAU variants addresses the material route, while process, substrate, channel, thermal-coupling or control claims can remain relevant independently of the material name. The configuration review therefore includes the drawings, bill of materials and manufacturing method as well as the sorbent composition. [T08]

The updated screen identifies BASF's air-conditioning and coated-exchanger family, NuMat's hybrid adsorbent-granule family, Framergy's amorphous-MOF family and Montana/AirJoule's thermally coupled contactor and water-harvesting family. The published claim features are mapped to the current product choices. These include channel arrangements, adsorbent/binder structures, amorphous processing and sealed or vacuum contactor architectures. The action will use the resulting claim charts to obtain the required professional country and configuration conclusions before affected commercial manufacture or sale. [T09–T12]

Savery's protection and third-party clearance are maintained separately. An owned system patent supports the company's competitive position, while the FTO process tests whether particular implementation choices encounter another party's enforceable rights. Where an issue is identified, the response can be a documented design adjustment, an authorised supply or licence arrangement, or a revised territory plan. The technical lead and IP adviser will review the adopted response before release and repeat the assessment when materials, geometry, suppliers, functionality or markets change.

International protection and commercial execution

European product development and initial manufacture remain the priority. Before expansion, the company will assess the national members and applicable claims of relevant families in the specific North American and Asian countries proposed for production or sale. Future protection will focus on the commercial jurisdictions and product improvements that justify the filing and prosecution cost. A PCT publication is an international filing route, not a worldwide granted right; the protection plan is implemented through the relevant national or regional procedures.

The scientific board supports disclosure and technical review. Norbert Stock contributes materials-science expertise and Aline Auroux contributes adsorption and thermodynamic measurement expertise. Scientific contribution and access to results do not by themselves assign ownership. Project and collaboration agreements will align publication review, invention disclosure and commercial use with each party's actual contribution. Frédéric Cui retains product and system responsibility, coordinating protection decisions with the manufacturing and deployment milestones.

The resulting IP strategy protects what makes the product reproducible and serviceable as well as what can be patented. It gives investors a visible chain from owned background rights to current configuration, controlled improvements, supplier transfer and country-specific market entry. The separate FTO status note records the present assessment and the next product-release decisions in the same terms.

2 Market

2.1 Market opportunity

Cooling continuity is a purchasing requirement

AirTransformer serves facilities that must keep cooling through heatwaves while operating within electrical and water limits. Our initial customers are data-centre owners and operators with an identifiable air-side cooling duty and access to suitable low-temperature heat. Their investment decision combines continuity of service, installed cost, annual operating expense and the ability to integrate equipment into a working facility. A cooling system that saves energy but increases the risk of service interruption will not meet this purchasing requirement.

This need is growing alongside digital infrastructure. The IEA projects global data-centre electricity consumption to rise from 485 TWh in 2025 to approximately 950 TWh in 2030. These figures describe whole facilities; AirTransformer addresses the cooling service allocated to its configuration. CBRE reports continuing European capacity growth alongside limited grid availability and construction delays. The European Commission also identifies cooling-water demand and electricity use as material sustainability issues for data centres. Together these trends create demand for equipment that can provide useful cooling with less electrical power and external make-up water. [M1–M3]

We initially address two purchasing situations. In an existing urban facility, the operator may need to add or retrofit a defined cooling duty without increasing its grid connection or external water requirement proportionately. In a new facility or extension, the designer can integrate available heat, heat rejection and air treatment before construction. The first route requires manageable disruption and compatibility with retained equipment. The second allows the interfaces to be designed together. Savery's qualification process covers both, while the existing cooling provision supports continuity during installation and commissioning.

The owner or developer approves investment, facilities engineers define interfaces, procurement allocates contractual responsibility and the operator accepts and maintains the installation. Savery provides the integrated equipment and application engineering needed to connect these decisions. Our offer states the assigned cooling capacity and outlet conditions, heat-source and sink requirements, auxiliary electricity, water balance, maintenance access and acceptance test. Data-centre liquid-cooling architectures can coexist with an assigned air-side duty; we do not present the product as a replacement for every cooling loop in a facility.

Site qualification is a joint engineering and commercial decision. Usable heat must be available at the required temperature, capacity and operating times. Heat rejection, air conditions, space, pressure drop and maintenance access must fit the installation. Water savings are assessed against the actual reference cooling and water-use arrangement. The same site must also offer an economic case strong enough to support procurement. Assessing these conditions together avoids counting a site as serviceable because it meets only one attractive criterion.

A quantified initial serviceable market

For commercial planning we define an initial annual serviceable market in 2031 covering urban colocation development in Frankfurt, London, Amsterdam, Paris and Dublin. These five markets provide a bounded entry geography across Germany, the United Kingdom, the Netherlands, France and Ireland. The planning case excludes North America, industrial process applications, campuses and the retrofit installed base. Those routes provide expansion potential as configurations and delivery capability mature.

CBRE's third-quarter 2025 report projected 624 MW of full-year new capacity across these five markets: Frankfurt 227 MW, London 193 MW, Amsterdam 31 MW, Paris 84 MW and Dublin 89 MW. This is a dated capacity forecast, rather than a list of AirTransformer purchase orders. We use a rounded 600 MW annual capacity flow as a flat planning reference for 2031, without assuming further growth. Our commercial case

allocates 10% to an air-side cooling duty and assumes that 25% of that duty jointly meets the heat-source, sink, climate, installation and economic requirements. These are Savery's planning assumptions, to be tested through account qualification. [M4]

The resulting 15 MW of serviceable annual cooling duty is valued at a planning installed price of €1,200/kW, giving an initial annual SAM of €18 million in 2031. The price case comprises €1,100/kW of pilot-stage hardware and €100/kW for installation and application engineering. It supports a capacity-based market estimate and customer-economic calculation; an individual offer will reflect its configuration and responsibilities. We do not infer a released physical machine rating from a 100 kW commercial planning equivalent. At joint qualification of 10–40%, the same flat-flow calculation gives €7.2–28.8 million. The range makes the significance of qualification explicit rather than implying that every new data-centre MW is addressable.

We target €0.6 million of sales within this same C05 geography and installed-service scope in 2029, €1.2 million in 2030 and €1.8 million in 2031. These correspond to 0.5, 1.0 and 1.5 MW of accepted cooling duty at the planning price. The 2031 target is 10% of the €18 million initial SAM. It requires product qualification, repeatable manufacture and delivery capacity, followed by customer conversion; it represents planned sales rather than secured orders. Maintenance revenue is assessed separately on contracted installed equipment and is excluded from both sides of this market-share comparison.

The wider opportunity includes suitable retrofits, further European markets and North American deployments, followed by qualified campus and industrial uses. Our company-wide growth model covers a broader product and geographic mix. Its €180.89 million revenue forecast for 2032 is therefore not used as the numerator against this narrowly defined C05 SAM. This separation allows the initial market-entry plan to be assessed on its own engineering, commercial and manufacturing assumptions.

Competition and the customer's alternative investment

Customers can retain compression cooling, improve economisation or water reuse, procure another desiccant or sorption system, or change their cooling architecture. AirTransformer must demonstrate value against the alternative the customer could actually install. The decision depends on useful cooling delivered at the interface, net electrical and thermal demand, external water withdrawal, space, capital cost and lifetime support.

AirJoule's current Prime disclosure describes water and dry-air production using low-grade waste heat, while its Core DH offering addresses industrial dehumidification with recovered water. The company reports a first full-scale Prime at Newark and a shipment for a European data-centre deployment planned for the fourth quarter of 2026. These disclosures show that low-temperature heat use alone is insufficient differentiation. Prime water-production metrics and Core DH humidity-control metrics also require their own comparison boundaries; they do not establish a matched AirTransformer cooling comparison. [M5]

Transaera addresses MOF-based outdoor-air treatment. Mojave offers liquid-desiccant air handling, including hydronic configurations. Munters provides established air-treatment and data-centre cooling alternatives. Atmospheric-water-harvesting suppliers address a related resource need. Our differentiation rests on the integrated air-side cooling service, compatibility with site heat and heat rejection, material and contactor engineering, whole-installation resource performance and accountable delivery. Productisation will make these differences testable through agreed performance maps and acceptance procedures. [M6–M8]

Savery retains product authority across application engineering, critical process specifications, assembly, control configuration and release. This provides a single technical responsibility for interfaces and performance, while qualified manufacturing and installation partners extend execution capacity. Batch records and field-service findings feed corrective action and controlled updates. For a customer deploying across several sites, repeatable commissioning, spares and maintenance are as important as the efficiency of an individual demonstration.

2.2 Business and revenue model and growth strategy

Equipment and service with a defined acceptance boundary

Our initial revenue model combines equipment, installation, application engineering and paid maintenance. The customer purchases a cooling duty delivered under agreed conditions, with corresponding thermal, electrical and water interfaces. Savery defines the supplied equipment and the customer or third-party infrastructure needed to complete the installation. For configurations that recover liquid water, the offer allocates collection, treatment, storage, metering and use qualification explicitly. The commercial agreement connects this scope to commissioning and acceptance criteria.

The company growth model uses a €120,000 reference package comprising €110,000 equipment, €8,000 installation and €2,000 application engineering. This is a reference commercial structure for the model. The capacity-based entry case in §2.1 separately uses €1,200/kW installed and a nominal 100 kW-equivalent contract value of €120,000. Actual rated capacity, redundancy, interfaces, water equipment and installation work determine a project quotation. Shared infrastructure and existing customer equipment are reconciled in the supply schedule before pricing. [M9]

Our pilot-stage hardware benchmark is €1,100/kW. The manufacturing development path targets €900/kW at the scale conditions established by procurement, process yield and qualified production capacity. The reduction is a scale-up objective, rather than a current unit price. It depends on reproducible material processing, standardised assembly, supplier qualification and lower rework, while preserving service performance. Commercial offers will therefore distinguish the product's current stage and configuration from future industrial cost targets.

Sales and application engineering jointly qualify the account before an order enters production. The technical proposal establishes the required duty, source and sink conditions, installation sequence, reference equipment and acceptance process. Procurement then agrees the supply boundary, delivery obligations and payment events. The operating model uses staged billing of 20% on order, 20% on manufacturing completion, 55% on acceptance and 5% after a six-month retention period. These planning terms connect cash collection to delivery progress, while retaining customer protection through commissioning and early operation. [M9]

Maintenance begins after the first year in the operating model, with an annual reference fee of 3% of equipment value, 60% attachment and 90% renewal. These assumptions apply to the corresponding contracted fleet. They are separate from initial support and warranty obligations, so the same service is not charged twice. Remote information, inspection, calibration, replaceable components and field response sustain the accepted operating envelope. Maintenance planning expands with the installed base, rather than treating service as an unresourced percentage of revenue.

Customer economics under explicit operating conditions

The purchasing case combines avoided electrical consumption, external make-up water and the cost of the useful regeneration heat, together with maintenance. A calculated example illustrates the economic conditions behind our 24–36-month payback objective. For an assigned 100 kW cooling duty operating 8,000 hours annually, useful cooling is 800 MWh/year. A reference electrical EER of 3 requires 266.7 MWh/year of electricity. A product electrical EER of 8–10 gives 80–100 MWh/year, reducing electricity by 166.7–186.7 MWh/year. These are scenario calculations using an engineering performance range, not a universal measured saving.

At €0.35/kWh of electricity, that reduction is worth approximately €58,300–65,300 annually. The example allows 800 MWh/year of regeneration heat at €0.01/kWh, costing €8,000. A reference external-water requirement of 1.5 litres per kWh of cooling and a 50% reduction give 600 m³/year of avoided withdrawal, worth €2,400 at €4/m³. After a further €2,000/year service allowance, net annual savings are approximately €50,700–57,700. A €120,000 installed investment then gives approximately 25–28 months' simple payback. The result lies within the 24–36-month objective under these stated assumptions.

We will calculate each customer's case using its actual load, operating hours, tariffs and source conditions. Lower electrical tariffs or utilisation, purchased heat with a higher opportunity cost, or additional integration work can extend payback. Retained cooling, heat rejection, water treatment and all auxiliaries enter the comparison. Water already recovered or reused by the reference system is credited consistently. Avoided grid or plant expansion is considered only against an identified alternative investment and is not counted a second time as operating savings.

Market entry and repeat deployment

The CETP project provides an established collaboration and site path with HPE for the progression towards TRL7. Our EIC action builds the manufacturing, reliability, compliance and independent qualification needed for a repeatable TRL8 product. Commercial entry follows that product qualification. The initial C05 sales case therefore begins in 2029, following the planned March 2029 completion of the 24-month action. Customer development and specification work can proceed earlier without assuming that a future product is already released.

Ina Wang will lead account development and customer requirements with the application and engineering team. A planned HVAC and data-centre business-development lead joins from project month 12 through the investment-funded deployment programme. This role adds sector procurement, integrator and technical-sales experience while retaining Ina's commercial and customer-interface responsibilities. It develops a qualified account funnel,

manages the conversion from assessment to pilot and repeat order, and builds installation and service channels. Manufacturing and service commitments follow product release, trained personnel and qualified partner capacity.

An accepted installation creates reusable interface specifications, commissioning methods, operating evidence and maintenance procedures. These support expansion at the same site and subsequent sites in a customer portfolio. Our initial annual C05 acceptance targets of 0.5, 1.0 and 1.5 MW are compatible with planned small-batch capacity of roughly one nominal 100 kW equivalent per month in 2029, rising towards two per month in 2030. These are capacity-planning equivalents, not a claim that every site uses the same physical machine. Booking includes an initial 16-week production and installation preparation allowance and retains capacity for rework and field support.

The broader company scenario rises from €0.48 million revenue and four accepted reference units in 2027 to €180.89 million and 1,498 units in 2032. This represents approximately 228% annual compound revenue growth over five years, or a multiplication by about 3.28 each year. Such growth requires substantial subsequent capital, manufacturing qualification, trained field teams, channel conversion and repeat purchases across the wider product and geographic portfolio. It is an ambitious company scaling scenario rather than the base case for this C05 entry market. [M9]

The growth model assumes each sales and application-engineering pair develops eight qualified accounts annually, 50% convert to first orders and the initial deployment averages three units. It retains a twelve-month contact and validation period before initial conversion. Planned industrial expansion moves from small-batch preparation to pilot-factory qualification and later replication. The EIC-supported product qualification provides an essential input to that trajectory, while investment funds the manufacturing and deployment capability. We will review the funnel, accepted capacity, installation productivity and cash requirements together before expanding commitments.

Equipment and maintenance provide the near-term revenue base. Cooling-as-a-Service and Energy-as-a-Service can follow once operating evidence, metering and maintenance establish a financeable service. Such contracts require an equipment owner, asset financing, performance and downtime responsibilities, and lifetime maintenance provision. The current equipment forecast does not add these future service revenues, water sales, licensing or OEM income on top of the same underlying equipment sales. This avoids attributing several revenue models to one asset before the commercial structure exists.

2.3 Broader impact

AirTransformer's intended climate-adaptation benefit is continued cooling when extreme heat coincides with constraints on electrical power or water. Lower electricity use at equal cooling service can preserve headroom within a facility's connection. Reduced external make-up demand can lessen vulnerability to water restrictions. HPE's site participation in CETP provides a real collaboration path, while EIC product qualification will establish the repeatable equipment, installation and maintenance needed to transfer the benefit between sites.

Our engineering objectives are 60–70% lower cooling electricity and 40–60% lower external water withdrawal in suitable configurations and comparison conditions. Product validation will report the reference equipment, delivered cooling, weather, heat supply and sink, all auxiliary loads, treatment and maintenance. Continuity will be assessed through compliant operating hours, unmet cooling and recovery after disturbances. Resource efficiency will use equivalent delivered cooling over the same evaluation period. The distinction preserves the practical value of maintaining service under a cap without confusing a service shortfall with an efficiency gain.

Recovered water can support qualified non-potable needs. External-water savings remain the principal water metric because internal consumption, evaporation, storage and discarded water determine what actually substitutes for supply. Treatment and intended use are assessed before release. The commercial case credits useful water within actual demand and does not require an unsupported daily water-production promise.

Lifecycle assessment will cover materials, manufacturing, transport, operation, maintenance, replacement and end of life. It will apply common service, lifetime and regional energy assumptions to the product and reference. The burden and opportunity cost of recovered heat enter the assessment according to its alternative use and allocation. Avoided refrigerant leakage or infrastructure expansion will be quantified only where the selected product and reference provide the corresponding evidence.

European economic value arises through engineering, critical material processing, contactor manufacture, assembly, testing, product release and lifecycle service. The project plans 258 person-months of execution effort across 14 roles. This is a project resource commitment, distinct from current headcount. Controlled process transfer, documented assembly and trained deputies support skilled work and reduce dependence on a single

individual. Subsequent jobs and expenditure will follow actual production and deployment, with their location recorded rather than inferred from company-wide international sales.

2.4 Challenge alignment

The project addresses the C05 priority of combating extreme heat in urban environments through urban cooling. Urban critical facilities require dependable cooling through increasingly demanding heat events. At data centres, interruption can affect the digital services used by businesses, public services and households. Our approach uses suitable available low-temperature heat to reduce the electrical and external-water requirements of an assigned cooling duty, supported by a verified operating envelope and customer acceptance. This connects the material and thermal engineering to a specific adaptation service. [M10]

C05's examples include several building and material approaches, while leaving room for other deep-tech urban-cooling solutions. Our product route is grounded in the existing TRL6 machine, Savery's IP and the funded CETP/HPE collaboration. The EIC action advances product manufacturing, reliability, compliance and deployment readiness. It will produce evidence that a prospective operator can use to decide whether the equipment fits its service, utility and installation constraints.

Hospitals, campuses and public buildings provide potential replication routes where heat sources, air-side cooling needs and operating responsibilities fit the qualified product. They are prospective markets and are not presented as secured sites or customers. Through Business Acceleration Services and the Adaptation Mission network, we will seek appropriate regional users and procurers, using performance maps, installation boundaries, resource comparisons and service responsibilities to support assessment. This is a dissemination and market-development route for a verified cooling product.

2.5 STEP impact

Savery will turn advanced-materials and system-engineering capability into equipment that European customers can procure and maintain. The project joins CAU processing and contactor quality with system assembly, control configuration, calibration and product release. Keeping these critical specifications and feedback processes under Savery's product authority supports European technological competitiveness in resource-efficient cooling.

The proposed European manufacturing route retains process recipes, inspection methods, software and test specifications needed to reproduce performance. Qualified partners can execute defined production steps and installations while batch records and release criteria preserve technical accountability. The supply strategy develops suitable alternative sources and requalifies substitutions against cyclic performance, durability and the operating envelope. Standard components can be globally sourced without dispersing responsibility for the product's critical functions.

This capability reduces exposure to a single manufacturing or service route and creates an additional European equipment option. We will measure the contribution through the location and expenditure of critical development, processing, assembly, testing and support, together with trained roles and qualified suppliers. A company-wide Europe-and-North-America revenue forecast does not by itself establish EU jobs or EU installed capacity. The STEP contribution follows the activities and products actually delivered within the EU market.

3 Team Financial Needs and Implementation

3.1 Team capability

Savery's operating core connects thermal engineering, equipment delivery and commercial requirements. Frédéric Cui leads system architecture, technical methods and product decisions. His experience includes building-energy modelling and experimental validation at ARMINES, followed by adsorption research and prototype development. This gives the project a technical authority able to connect material behaviour, heat and mass transfer, interfaces and machine performance.

Qin Yu leads engineering execution, builds, suppliers and commissioning preparation. His documented heat-pump installation and technical-project experience supports the translation from design to practical equipment and site work. Ina Wang leads commercial requirements, account development and customer acceptance interfaces, drawing on her B2B logistics, agri-food and pricing experience. The project supplements these capabilities with dedicated productisation, controls, quality, reliability and field resources, rather than expecting the core to cover every specialist task.

Norbert Stock and Aline Auroux are members of Savery's scientific board. Stock contributes materials-science expertise to the CAU route, processing windows and interpretation of material performance. Auroux contributes adsorption and thermodynamic-measurement expertise, supporting the interpretation of heat, sorption and

qualification results. Their scientific-board contribution is distinct from paid execution effort. The budget allocates no employee person-months automatically to a board title.

Core team and planned commitment

Name	Gender	Founder	Position and department	Key competences	Project commitment
Frédéric Cui	Male	Yes	Operational CEO and system technology	Thermal modelling, sorption machines, experimental validation and product architecture	50%
Qin Yu	Male	No	Engineering and production lead	HVAC installation, supplier coordination, technical projects and commissioning	87.5%
Ina Wang	Female	No	Commercial lead	B2B development, pricing, customer requirements and acceptance interfaces	50%

The commitment percentages use the same 24-month full-time denominator and represent the planned EIC action allocation. Frédéric is the founder. Scientific-board contributions and future hires are shown separately.

Responsibilities and planned execution effort

The 24-month action provides 258 person-months across 14 execution roles. Named core allocations are 12 PM for Frédéric, 21 PM for Qin and 12 PM for Ina. Their planned project commitments are respectively 50%, 87.5% and 50% of one full-time role over the action. These are planned project allocations with a common 24-month denominator. They are separate from historical descriptions of full-time venture involvement and from legal or equity status.

Responsibility	Accountable lead and execution resource	Planned entry and effort	Productisation result
Architecture and technical decisions	Frédéric Cui	M1–24; 12 PM	Requirements, configuration decisions and release authority
Builds and supplier execution	Qin Yu	M1–24; 21 PM	Build records, suppliers, assembly and commissioning preparation
Customer requirements and acceptance	Ina Wang	M1–24; 12 PM	Qualified accounts, site requirements and acceptance interfaces
Thermal modelling	Planned modelling specialist under Frédéric	M1–18; 18 PM	Product models and engineering verification
Supervisory controls	Planned controls specialist under Frédéric	M3–20; 18 PM	Constrained control, comparison and fallback validation
Process transfer	Planned process specialist under Qin	M1–24; 24 PM	Manufacturing windows and repeatable contactors
Reliability and independent-test preparation	Planned reliability engineer	M1–24; 24 PM	Durability, contamination, fault and repeat-test evidence
Experimental execution	Planned test technician A	M1–24; 24 PM	Instrumentation, test operation and records
Field engineering	Planned field engineer under Qin	M7–24; 18 PM	Site integration, commissioning and service handover

Responsibility	Accountable lead and execution resource	Planned entry and effort	Productisation result
Data and project records	Planned data and project resource	M1–24; 12 PM	Traceable records, data management and reporting
Design for manufacture	Planned DfM engineer under Qin	M1–21; 21 PM	Assembly, repair and production documentation
Quality and compliance	Planned quality and compliance engineer	M4–21; 18 PM	Quality system, applicable compliance and technical file
Qualification test capacity	Planned test technician B	M4–21; 18 PM	Parallel qualification and rework capacity
Embedded software	Planned embedded-software engineer	M4–21; 18 PM	Deployable controls, diagnostics and software release

The planned specialists represent funded roles to recruit or secure through appropriately scoped support; they are not presented as employees already in post. Frédéric owns technical specifications and the interaction between modelling and controls. Qin owns manufacturability, supplier interfaces and build readiness. The reliability and quality resources prepare evidence for independent testing and release. The field engineer joins early enough to survey interfaces and prepare commissioning before equipment reaches the site. Commercial use follows the released configuration and its operating boundary.

A planned HVAC and data-centre business-development lead starts in M12 under the separate investment programme, with 25 PM through M36 and a €212,500 employer-cost provision. The role strengthens sector-specific sales, integrator channels and tender conversion. A service application engineer and field technicians subsequently expand support for the installed fleet. These commercial deployment resources sit outside the 258 grant-funded PM, preventing commercial expansion from consuming the same staff capacity twice.

Recruitment follows the sequence of technical dependencies. Process, reliability and modelling capacity enters at the start; supervisory-control expertise follows in M3; quality, additional test capacity and embedded software enter in M4; field engineering enters in M7. Hiring specifications require demonstrated competence in the assigned deliverable. Scoped specialist services provide a bridge where recruitment lead times threaten a gate, with Savery retaining technical ownership and budget control. No cost is charged twice where CETP and EIC contributors share technical knowledge or equipment.

Governance motivation and continuity

Frédéric makes architecture and product decisions, supported by scientific-board review in its specialist areas. Qin prepares manufacturing, integration and commissioning evidence. The reliability and quality leads organise testing and compliance inputs so that a delivery deadline does not override a safety or qualification result. Ina represents customer requirements and makes acceptance and service obligations visible before commitments. Financial approvals follow the company's authorised signatory arrangements.

Monthly reviews combine progress, staffing, supplier readiness, technical risk and cash requirements. Each gate has a defined evidence package and an accountable technical decision. External testing and counsel provide specialist assessment within their contracted scope. Site partners contribute the agreed interfaces and access, while Savery remains responsible for the product and its supplied service. CETP knowledge and HPE site collaboration support the route without turning a future CETP deliverable into an unexamined EIC dependency.

Documented designs, software versions, test procedures, supplier instructions and change decisions allow colleagues to continue work when the founder is unavailable. Cross-training gives deputies for build, test and customer interfaces. Role-appropriate remuneration, technical development and ownership of product outcomes support retention. Any equity incentives follow the company's actual approvals and rights records; individual project effort is not used as a proxy for shareholding.

Ownership and incentives

CURMA SOLUTIONS holds Savery Tech's issued shares. The signed shareholder declaration of 7 April 2026 records 100% of issued capital and voting rights; current investor arrangements are BSA-AIR instruments that have not converted or diluted those issued holdings. Conversion will use the valuation established at the next financing. A 10% option pool is reserved for key team members. An illustrative allocation including only the

reserved pool is 90% for the existing holder and 10% for the pool, before BSA-AIR conversion and new financing. The actual fully diluted ownership schedule will follow the conversion terms, valuation and option arrangements. Frédéric's indirect ownership through CURMA is not counted as a second direct Savery holding.

Gender equality throughout the organisation

Our measures cover scientific, engineering, operations, commercial, administrative and management roles. Ina will coordinate implementation with Frédéric's accountability for recruitment and progression. Every new role will use inclusive job descriptions, structured skills criteria and consistent interview scoring. Recruitment channels will include networks that reach women in thermal engineering, materials, controls and production. Candidate screening and shortlists will be reviewed for unjustified differences, including in technician and other non-management hiring.

Pay and progression reviews will compare work of equivalent responsibility. Project scheduling will support predictable hours, reasonable flexibility and return after caring leave where compatible with safe site and laboratory operation. Protective equipment and workplace arrangements will fit the workforce rather than assuming a single user profile. A clear conduct and reporting procedure will provide a route for concerns, with confidential handling and action by management. Training and access to technically significant tasks will be reviewed across the organisation.

Quarterly reviews will track recruitment, hiring, retention, training and progression by role family, using voluntary personal information and appropriate privacy safeguards. The review will distinguish the scientific board, current staff and planned recruits, avoiding a headline management percentage that conceals the rest of the organisation. Actions will address the point in the employment process where imbalance occurs, while preserving skill and safety requirements.

3.2 Risk level of the investment and leverage effect

Savery requests €2,485,583.33 in grant and €2,000,000 in EIC investment, a combined EIC request of €4,485,583.33. The 24-month grant action has estimated eligible costs of €3,550,833.33, with a €1,065,250.00 non-grant share. The investment forms part of the company's €8 million Seed target, alongside €6 million from other sources. It supports productisation and deployment through distinct funding and decision processes.

The funding need follows the industrial transition from a TRL6 machine to a reproducible and supportable product. Grant work addresses design for manufacture, CAU processing and batch quality, reliability, engineered control, applicable compliance, independent verification and TRL8 qualification. Investors must additionally support the manufacturing and service capacity, commercial conversion and working capital required to deploy equipment. A technically successful prototype does not by itself fund these obligations or remove the risk of a slow first procurement cycle.

At application preparation, Savery reports €300,000 cash and no outstanding debt or short-term liabilities. Investor financing uses unconverted BSA-AIR instruments, with conversion at the valuation of the next financing. These current resources support mobilisation alongside the planned round. Our action cash plan totals €3,908,916.67 because it includes full asset purchases of €529,000, while eligible costs include €170,916.67 of action depreciation. The residual cash requirement after grant is €1,423,333.33. This amount already includes the non-grant share and the purchase-to-depreciation difference; it is not added to the Seed target as a second financing round. The grant request and its payment timing remain distinct from investment proceeds and customer collections.

The separate investment and commercial-deployment work programme totals €2,112,500. It covers production tooling, manufacturing-partner readiness, customer and channel development, first installations, service capability and initial inventory and receivables. The EIC investment use plan assigns €1.9 million to these deployment activities and €100,000 to action co-financing. Other Seed sources cover the €212,500 technical-business-development role and the remaining action cash requirement, with the wider company's financing uses reconciled in its funding plan. These allocations identify intended uses once funding is available and avoid assigning the same euro twice.

Industrialisation carries coupled risks: durability and contamination may affect service performance; batch yield or supplier changes may affect released capacity; independent testing may require redesign; site commissioning and customer acceptance may take longer than planned. Early commercial customers also expect credible maintenance and response capability. These risks require expenditure before dependable sales cash can be established. EIC support can help bridge this period and support diligence by private investors whose commitment depends on demonstrable product and market progress.

The funded CETP programme provides an existing collaboration and TRL7 site path. Its activities and costs remain distinct from the EIC product-qualification action. Existing know-how and data reduce development risk; EIC resources address the additional manufacturing, reliability, compliance and deployment evidence. The EIC schedule retains its own qualification route through March 2029 so that completion does not depend on later CETP results without an alternative test route.

We will organise investor diligence around product, manufacturing and commercial gates. During the first six months, investors can assess the released requirements, ownership position, cost build and manufacturing plan. Subsequent gates add repeatable batches, reliability and independent-test evidence, followed by product qualification and accepted customer service. The company seeks the remaining Seed sources alongside this evidence programme and within the investment mobilisation period. Existing investor letters of intent can support the account of interest according to their actual terms and dates; requests, conditional intentions and funding already received are recorded separately.

Deployment commitments will follow available funds, qualified production slots and customer obligations. Staged customer payments can support working capital, while inventory, acceptance delays and retention must still be financed. The plan therefore carries specific inventory and receivables provision rather than assuming that an order deposit removes the cash need. If financing or acceptance slips, the company can sequence purchases, manufacturing slots and commercial expansion while preserving essential safety and product-qualification work. This makes the leverage sought from EIC support concrete: a qualified product and a financeable route to manufacturing and repeat deployment.

3.3 Implementation and risk management

Work plan and resources

The 24-month grant action plans April 2027–March 2029 and 258 person-months across fourteen execution roles. Five work packages move the current TRL6 complete machine through product requirements, manufacturing transfer, reliability and industrial controls, independent product testing and reproducible deployment to TRL8 qualification. Frédéric owns product decisions, Qin engineering and release, and Ina application requirements and acceptance. Planned specialists have assigned entry months and effort for thermal validation, supervisory controls, embedded software, DfM, process quality, reliability, test operation, field engineering and data coordination. Scientific board contributions remain separate from paid execution PM.

WP	Months	PM	Eligible EUR
Product requirements and engineering baseline	1–6	23.000	282,250.00
Manufacturing transfer and batch qualification	1–18	71.250	911,125.00
Reliability controls and conformity validation	3–21	73.375	1,210,989.58
Product qualification and reproducible deployment	6–24	76.875	1,006,781.25
Project management data IP and exploitation	1–24	13.500	139,687.50

The action qualifies six manufacturing batches and twelve reusable contactors, with seventy-two external sorption/thermal analyses and eighteen destructive witness examinations. A planned 4,000-hour internal reliability programme and 1,200 independent fixture-hours address ageing, contamination, maintenance and failure recovery. The industrial qualification unit completes sixty climate/load days in M10–M14 before field installation. Eight months of incremental field qualification plan 1,600 operating hours, use-specific water monitoring and an acceptance dossier at M24. Requirements, baseline configuration, uncertainty and acceptance methods are set by M6; performance claims follow actual measurement conditions and results.

The itemised eligible budget is €3,550,833.33, including €1,648,500.00 personnel, €325,000 independent subcontract tasks, €51,750 travel, €170,916.67 equipment depreciation, €709,500 goods/services and €645,166.67 indirect costs. The grant request is €2,485,583.33 at 70%, with €1,065,250.00 co-financing. Capital purchase cash of €529,000 is distinguished from eligible depreciation. The selected hybrid uses existing facilities, limited process/assembly fixtures and purchased independent tests; a general new synthesis laboratory is unnecessary.

CETP already funds the 2026–2029 research/HPE route towards TRL7. EIC uses available research and data while financing different industrial-design, batch-quality, reliability, conformity and product-qualification outputs. Separate employee-month, equipment-use, test and invoice records prevent duplicate charging. Independent EIC facilities and the existing machine avoid dependency on CETP results delivered after March 2029.

The €2m EIC investment request remains within the €8m Seed target, alongside €6m other financing. It allocates €1.9m to the investment work package and €100,000 to innovation co-financing. WP6 plans serial-production readiness, commercial service recruitment, initial installation support and working capital, outside grant costs. A planned HVAC/data-centre business-development lead starts at M12. Product qualification at M24 precedes initial commercial deliveries and the capacity ramp supporting five, ten and fifteen nominal 100kW-equivalent sales in 2029–2031. These are scoped commercial assumptions rather than existing machine ratings or orders.

Technical financial and delivery risks

The product programme uses evidence gates to control commitments. Frédéric holds product authority, Qin coordinates engineering and suppliers, and the quality and reliability leads prepare independent-test and release evidence. Monthly reviews combine technical results, workforce capacity and cash commitments. An unresolved performance or safety deviation triggers a defined technical response before release.

Risk	Consequence and response	Decision owner and timing
A rating point fails under hot or humid conditions	Restrict the qualified operating domain, adjust contactor or thermal design and repeat the complete-service comparison. The retained cooling installation supplies duties beyond the released domain.	Frédéric and thermal engineer; specification review M6 and each performance campaign
Batch variation or coating degradation affects output	Use traceable lots, after-shaping release tests and contamination or ageing checks. Quarantine nonconforming material and qualify any process or supplier change.	Process and quality leads; batch gates M6–18
Predictive or learning control behaves poorly outside its training domain	Independent physical limits, sensor diagnostics, operator override and deterministic fallback retain authority. Validate disturbances and recovery on reserved operating conditions.	Controls and embedded-software leads; M9–20
Independent tests identify a conformity or reliability problem	Reserve time and resources for diagnosis, corrective design and repeat tests. Product release requires the applicable technical file and accepted evidence.	Quality and reliability leads; M9–24
Recruitment supplier or site readiness delays a gate	Use staged recruitment, a documented deputy and scoped specialist support. Reserve test capacity early and retain an independent qualification route when field access is constrained.	Qin and project coordinator; monthly and before each purchase or test commitment
CETP and EIC activities use the same resource	Record exclusive employee time, equipment use, test purpose and invoices by project. Reuse technical information with its access rights while assigning each cost once.	Project coordinator and accountant; monthly cost review
Investment or grant receipts arrive later than the cash scenario	Phase procurement and deployment against available funds. Maintain the delayed-receipt case and protect the essential product-qualification sequence. Investment requests and future company rounds are conditional financing plans.	Frédéric and finance support; before commitments and at each funding gate
Third-party rights affect a released configuration or territory	Complete product and claim comparisons, obtain appropriate rights or change the affected design or market route. Reassess changes in materials, geometry, controls and supply.	Frédéric with specialist counsel; before the affected commercial release

The investment programme follows product and manufacturing gates rather than overriding them. Grant work establishes the qualified configuration and its evidence. Commercial tooling, fleet service and working capital then support repeat deployment within the funding available. A delay in a later CETP result does not extend the EIC critical path beyond the planned March 2029 completion.

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